

Indeterminate forms

① $\frac{f(x)}{g(x)} \left(\frac{0}{0}, \frac{\infty}{\infty} \right) \rightarrow$ use L'Hospital Rule directly

② $f(x) \cdot g(x) [0 \cdot \infty] \rightarrow f(x) \cdot g(x) = \frac{f(x)}{1/g(x)}$

or

$$= \frac{g(x)}{1/f(x)}$$

↓
then use L'Hospital

③ $f(x)^{g(x)} [1^\infty, 0^0, \infty^0]$

↓ use $\ln / \log$ to

transform to a product / fraction

↓
use L'Hospital's rule

Note: If $\ln f(x)^{g(x)} \rightarrow a$

then $f(x)^{g(x)} \rightarrow e^a$

OR: If $\lim \ln f(x)^{g(x)} = a$

$$\Rightarrow \lim f(x)^{g(x)} = e^a$$

Example

$$\textcircled{1} I = \lim_{x \rightarrow 0} (1+x)^{1/x} \quad [1^\infty]$$

we have:

$$\lim_{x \rightarrow 0} \ln \left[(1+x)^{1/x} \right]$$

$$[\text{note: } \ln a^b = b \cdot \ln a]$$

$$= \lim_{x \rightarrow 0} \frac{1}{x} \cdot \ln(1+x)$$

$$= \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} \quad \left[\frac{0}{0} \right]$$

L'Hospital

$$\rightarrow \lim_{x \rightarrow 0} \frac{[\ln(x+1)]'}{(x)'} =$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{x+1}}{1} = \lim_{x \rightarrow 0} \frac{1}{x+1} = 1$$

$$\Rightarrow \lim_{x \rightarrow 0} \ln I = 1 \Rightarrow \lim_{x \rightarrow 0} I = e^1 = e$$

④ Indeterminate form : $f(x) - g(x)$ $[\infty - \infty]$

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$$

$$= \lim_{x \rightarrow 0^+} \frac{\sin x - x}{x \cdot \sin x}$$

L'Hospital

$$\downarrow$$
$$= \lim_{x \rightarrow 0} \frac{(\sin x - x)'}{(x \cdot \sin x)'} \quad \leftarrow \text{product rule}$$

$$= \lim_{x \rightarrow 0} \frac{\cos x - 1}{1 \cdot \sin x + x \cdot \cos x}$$

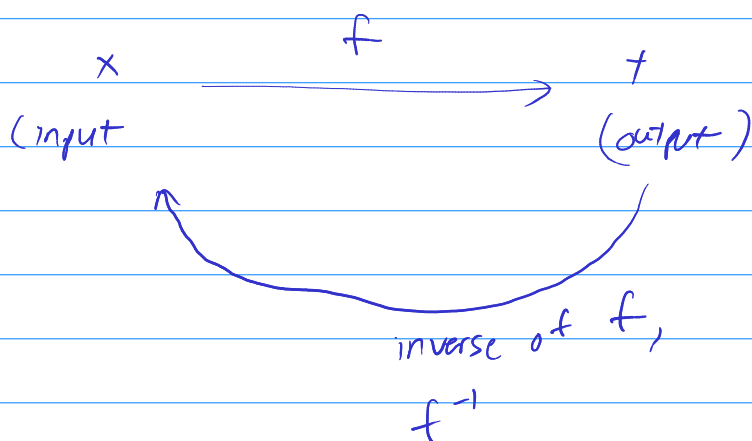
$$= \lim_{x \rightarrow 0} \frac{(\cos x - 1)'}{(\sin x + x \cdot \cos x)'} \quad \leftarrow \text{product rule}$$

$$= \lim_{x \rightarrow 0} \frac{-\sin x}{\cos x - x \cdot \sin x + 1 \cdot \cos x}$$

$$= \frac{0}{2} = 0$$

6.7. Derivatives and Integrals involving trigonometric Functions

⊗ Inverse functions



⊗ Derivatives of the inverse trigs.

$$\frac{d}{dx}[\sin^{-1}u] = \frac{1}{\sqrt{1-u^2}} \frac{du}{dx}$$

$$\frac{d}{dx}[\cos^{-1}u] = -\frac{1}{\sqrt{1-u^2}} \frac{du}{dx}$$

$$\frac{d}{dx}[\tan^{-1}u] = \frac{1}{1+u^2} \frac{du}{dx}$$

$$\frac{d}{dx}[\cot^{-1}u] = -\frac{1}{1+u^2} \frac{du}{dx}$$

$$\frac{d}{dx}[\sec^{-1}u] = \frac{1}{|u|\sqrt{u^2-1}} \frac{du}{dx}$$

$$\frac{d}{dx}[\csc^{-1}u] = -\frac{1}{|u|\sqrt{u^2-1}} \frac{du}{dx}$$

Example : Find dy/dx

$$y = \sin^{-1}(1/x)$$

Chain rule: $u = \frac{1}{x} \Rightarrow y = \sin^{-1}(u)$

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= \frac{1}{\sqrt{1-u^2}} \cdot \left(\frac{1}{x}\right)'$$

$$= \frac{1}{\sqrt{1-(1/x)^2}} \cdot \left(-\frac{1}{x^2}\right)$$

[note: $\left(\frac{1}{x}\right)' = (x^{-1})'$

$$= -1 \cdot x^{-2}$$

$$= -\frac{1}{x^2}]$$